Online Examination Report

Date:	13.08.2014
Attempts:	1
Examinee:	Mwakagile, Faraja
Date of birth:	7/27/1979
Subject:	020-AGK
Result:	32,49 %
Time used:	00:01:00 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	3639	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
2	2876	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
3	2414	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
4	2465	21.2.3	Aeroplane: Wings, tail and control surfaces	1,00	1,00
5	97	21.3.1	Hydro-mechanics: basic principles	1,00	1,00
6	1229	21.4	LANDING GEAR, WHEELS, TYRES, BRAKES	1,00	1,00
7	556	21.6.3	Aeroplane: Pressurisation and air conditioning system	1,00	1,00
8	854	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
9	1763	21.8.2.2	Design, operation, system components, etc	1,00	1,00
10	2256	21.9.1	General, definitions, basic applications: etc	1,00	1,00
11	2340	21.9.2	Batteries	1,00	1,00
12	319	21.9.4.3	AC distribution	0,00	1,00
13	1885	21.10.8	Mixture	0,00	1,00
14	671	21.10.10	Performance and engine handling	0,00	1,00
15	2432	21.10.10.1	Performance	0,00	1,00
16	2470	21.10.10.1	Performance	0,00	1,00
17	2363	21.11.1	Basic principles	0,00	1,00
18	1363	21.11.1.2	Design, types of turbine engines, components	0,00	1,00
19	2174	21.12.2.2	Fire detection	0,00	1,00
20	1601	21.13	OXYGEN SYSTEMS	1,00	1,00
21	3100	22.	INSTRUMENTATION	0,00	1,00
22	2793	22.	INSTRUMENTATION	0,00	1,00
23	3094	22.	INSTRUMENTATION	0,00	1,00
24	3405	22.	INSTRUMENTATION	0,00	1,00
25	34	22.	INSTRUMENTATION	0,00	1,00
26	4130	22.	INSTRUMENTATION	0,00	1,00
27	1110	22.2	MEASUREMENT OF AIR DATA PARAMETERS	0,00	1,00
28	813	22.2.1	Pressure measurement	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	1884	22.2.1.1	Definitions	1,00	1,00
30	1910	22.2.4	Altimeter	1,00	1,00
31	1409	22.2.7	Mach meter	0,00	1,00
32	1800	22.3	MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	0,00	1,00
33	1139	22.3.3	Direct Reading Magnetic Compass	0,00	1,00
34	1595	22.4	GYROSCOPIC INSTRUMENTS	0,00	1,00
35	1459	22.4.1	Gyroscope: basic principles	0,00	1,00
36	1614	22.4.1	Gyroscope: basic principles	0,00	1,00
37	1711	22.4.2	Rate of turn and slip indicator/Turn Co-ordinator	0,00	1,00
38	226	22.4.5	Remote reading compass systems	0,00	1,00
39	884	22.6.3	Flight Director : design and operation.	1,00	1,00
40	852	22.8.1	Trim systems : design and operation.	1,00	1,00
			Total:32%	**13,00**	**40,00**

Attachment 2: List of result by topic

Licence
Date: 13.08.2014
Examinee: Mwakagile, Faraja
Location: Uganda Civil Aviation Authority

Your Result: 13,00 from 40,00 Points (32,50 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	20	9,00	20,00	45,00
21.2. AIRFRAME	2	1,00	2,00	50,00
21.3. HYDRAULICS	1	1,00	1,00	100,00
21.4. LANDING GEAR, WHEELS, TYRES, BRAKES	1	1,00	1,00	100,00
21.6. PNEUMATICS - PRESSU AND AIRCO SYSTEMS	2	1,00	2,00	50,00
21.8. FUEL SYSTEM	1	1,00	1,00	100,00
21.9. ELECTRICS	3	2,00	3,00	66,67
21.10. PISTON ENGINES	4	0,00	4,00	0,00
21.11. TURBINE ENGINES	2	0,00	2,00	0,00
21.12. PROTECTION AND DETECTION SYSTEMS	1	0,00	1,00	0,00
21.13. OXYGEN SYSTEMS	1	1,00	1,00	100,00
22. INSTRUMENTATION	20	4,00	20,00	20,00
22.2. MEASUREMENT OF AIR DATA PARAMETERS	5	2,00	5,00	40,00
22.3. MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	2	0,00	2,00	0,00
22.4. GYROSCOPIC INSTRUMENTS	5	0,00	5,00	0,00
22.6. AEROPLANE : AUTOMATIC FLIGHT CONTROL SYSTEMS	1	1,00	1,00	100,00
22.8. TRIM, Y/D, FLIGHT ENVELOPE PROTECTION	1	1,00	1,00	100,00
Total	**40**	**13,00**	**40,00**	**32,50**

1

Fuel enters a two-cycle engine through an

- [x] intake port and intake valve.
- [ ] intake valve and reed valve.
- [ ] intake port and reed valve.

ID:	3639(LTS:223000, LTM:1147)	Max. Points	1,00
Syllabus reference:	21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	Points	1,00

2

What should help prevent aircraft induced oscillation on a gyroplane?

- ☐ **Increasing cyclic control sensitivity.**
- ☐ **Adding a horizontal stabilizer.**
- ☒ **Lowering the center of gravity below the thrust line.**

ID:	2876(LTS:222981, LTM:1107)	Max. Points	1,00
Syllabus reference:	21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	Points	0,00

3

What effect does an increase in airspeed have on a coordinated turn while maintaining a constant angle of bank and altitude?

- ☒ **The rate of turn will increase resulting in an increased load factor.**
- ☐ **The rate of turn will decrease resulting in a decreased load factor.**
- ☐ **The rate of turn will decrease resulting in no changes in load factor.**

ID: 2414(LTS:222856, LTM:993)
Syllabus reference: 21.2.3 Aeroplane: Wings, tail and control surfaces

Max. Points 1,00
Points 0,00

4

At which speed will increasing the pitch attitude cause an airplane to climb?

- [x] **High speed.**
- [ ] **Any speed.**
- [ ] **Low speed.**

ID:	2465(LTS:222907, LTM:1044)	Max. Points	1,00
Syllabus reference:	21.2.3 Aeroplane: Wings, tail and control surfaces	Points	1,00

5

In addition to energy storage, the accumulator of the hydraulic system is used :

- [x] **for damping pressure surges in the system.**
- [ ] **as a pressure relief valve.**
- [ ] **for pressure storage.**
- [ ] **for fluid storage.**

ID:	97(LTS:222266, LTM:22)	Max. Points	1,00
Syllabus reference:	21.3.1 Hydro-mechanics: basic principles	Points	1,00

6

The illumination of the green landing gear light indicates that the landing gear is :

- ☐ locked-down and its door is locked.
- ☒ locked-down.
- ☐ in the required position.
- ☐ not in the required position.

ID:	1229(LTS:222522, LTM:278)	Max. Points	1,00
Syllabus reference:	21.4 LANDING GEAR, WHEELS, TYRES, BRAKES	Points	1,00

7

"Conditioned" air is air that has :

- ☐ had any moisture removed from it.
- ☐ had the oxygen content reduced.
- ☐ had the oxygen content increased.
- ☒ been controlled in respect of temperature and pressure.

ID:	556(LTS:222363, LTM:120)	Max. Points	1,00
Syllabus reference:	21.6.3 Aeroplane: Pressurisation and air conditioning system	Points	1,00

8

The pneumatic system accumulator is useful :

- ☒ **in emergency cases.**
- ☐ **to eliminate the fluid flow variations.**
- ☐ **to eliminate the fluid pressure variations.**
- ☐ **to offset for the starting of some devices.**

ID:	854(LTS:222428, LTM:184)	Max. Points	1,00
Syllabus reference:	21.6.3 Aeroplane: Pressurisation and air conditioning system	Points	0,00

9

The cross-feed fuel system enables:

- ☐ the supply of the jet engines mounted on a wing from any fuel tank within that wing.
- ☐ only the transfer of fuel from the centre tank to the wing tanks.
- ☒ the supply of any jet engine from any fuel tank.
- ☐ the supply of the outboard jet engines from any outboard

ID:	1763(LTS:222672, LTM:429)	Max. Points	1,00
Syllabus reference:	21.8.2.2 Design, operation, system components, etc	Points	1,00

10

Fuses are rated to a value by :

- ☐ **the number of volts they will pass.**
- ☐ **their resistance measured in ohms.**
- ☐ **their wattage.**
- ☒ **the number of amperes they will carry.**

ID: 2256(LTS:222793, LTM:930)
Syllabus reference: 21.9.1 General, definitions, basic applications: etc

Max. Points 1,00
Points 1,00

11

Immediately after starting engine(s) with no other electrical services switched on, an ammeter showing a high charge rate to the battery :

- ☐ **indicates a faulty reverse current relay.**
- ☒ **would be normal and is only cause for concern if the high charge rate persists.**
- ☐ **indicates a generator failure, thus requiring the engine to be shut down immediately.**
- ☐ **indicates a battery failure since there should be no immediate charge.**

ID:	2340(LTS:222829, LTM:955)	Max. Points	1,00
Syllabus reference:	21.9.2 Batteries	Points	1,00

12

The emergency lighting system must be able to function and supply a certain level of lighting after the main electric power system has been cut off for at least:

- ☐ **5 minutes**
- ☐ **10 minutes**
- ☐ **30 minutes**
- ☒ **90 seconds**

ID:	319(LTS:222313, LTM:69)	Max. Points	1,00
Syllabus reference:	21.9.4.3 AC distribution	Points	0,00

13

The octane rating of a fuel characterises the :

☐ **fuel electrical conductivity**

☒ **fuel volatility**

☐ **quantity of heat generated by its combustion**

☐ **the anti-knock capability**

ID:	1885(LTS:222720, LTM:468)	Max. Points	1,00
Syllabus reference:	21.10.8 Mixture	Points	0,00

14

Which statement is true concerning the effect of the application of carburettor heat?

- ☐ it reduces the density of air entering the carburettor, thus leaning the fuel/air mixture
- ☐ it reduces the density of air entering the carburettor, thus enriching the fuel/air mixture
- ☐ it reduces the volume of air entering the carburettor,thus leaning the fuel/air mixture
- ☒ it reduces the volume of air entering the carburettor,thus enriching the fuel/air mixture

ID:	671(LTS:222387, LTM:143)	Max. Points	1,00
Syllabus reference:	21.10.10 Performance and engine handling	Points	0,00

15

Under what condition should stalls never be practiced in a twin-engine airplane?

- [x] **With full flaps and gear extended.**
- [ ] **With climb power on.**
- [ ] **With one engine inoperative.**

ID: 2432(LTS:222874, LTM:1011)
Syllabus reference: 21.10.10.1 Performance
Max. Points 1,00
Points 0,00

16

How does V(NE) speed vary with altitude?

☒ **Remains the same at all altitudes.**

☐ **Varies inversely with altitude.**

☐ **Varies directly with altitude.**

ID:	2470(LTS:222912, LTM:1049)	Max. Points	1,00
Syllabus reference:	21.10.10.1 Performance	Points	0,00

17

The Engine Pressure Ratio (EPR) is the ratio of:

- ☐ **the total turbine outlet pressure to the total compressor outlet pressure.**
- ☐ **the total turbine outlet pressure to the total compressor inlet pressure.**
- ☒ **the total turbine inlet pressure to the total compressor outlet pressure.**
- ☐ **the total turbine inlet pressure to the total compressor inlet pressure.**

ID:	2363(LTS:222841, LTM:967)	Max. Points	1,00
Syllabus reference:	21.11.1 Basic principles	Points	0,00

18

A turbofan engine with an inlet air mass flow of 300 kg/s and a bypass mass flow of 250 kg/s has a bypass ratio of:

- ☒ *1.2*
- ☐ *6*
- ☐ *5*
- ☐ *2.2*

ID:	1363(LTS:222556, LTM:312)	Max. Points	1,00
Syllabus reference:	21.11.1.2 Design, types of turbine engines, components	Points	0,00

19

A gaseous sensor/responder tube fire sensor is tested by

☐ **checking the continuity of the system with a test switch.**

☒ **checking the wiring harness for faults but not the sensor.**

☐ **heating up the sensor with test power connection.**

☐ **checking the sensor with pressurized gas.**

ID:	2174(LTS:222769, LTM:906)	Max. Points	1,00
Syllabus reference:	21.12.2.2 Fire detection	Points	0,00

20

The state in which the breathing oxygen for the cockpit of jet transport aeroplanes is stored is :

- [x] **Gaseous.**
- [ ] **Chemical compound.**
- [ ] **Gaseous or chemical compound..**
- [ ] **Liquid.**

ID: 1601(LTS:222619, LTM:376)
Syllabus reference: 21.13 OXYGEN SYSTEMS

Max. Points 1,00
Points 1,00

21

Identify the runway distance remaining markers.

☒ **Yellow marker laterally placed across the runway with signs on the side denoting distance to end.**

☐ **Red markers laterally placed across the runway at 3,000 feet from the end.**

☐ **Signs with increments of 1,000 feet distance remaining.**

ID:	3100(LTS:222994, LTM:1131)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

22

In the Northern Hemisphere, if an aircraft is accelerated or decelerated, the magnetic compass will normally indicate

- ☐ a turn momentarily, with changes in airspeed on any heading.
- ☐ correctly when on a north or south heading while either accelerating or decelerating.
- ☒ a turn toward the south while accelerating on a west heading.

ID: 2793(LTS:222965, LTM:1091)
Syllabus reference: 22. INSTRUMENTATION

Max. Points 1,00
Points 0,00

23

How can a pilot identify a military airport at night?

- ☐ **White and red beacon light with dual flash of the white.**
- ☐ **Green and white beacon light with dual flash of the white.**
- ☒ **Green, yellow, and white beacon light.**

ID:	3094(LTS:222992, LTM:1129)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

24

How will the airspeed indicator react if the ram air input to the pitot head is blocked by ice, but the drain hole and static port are not?

- ☐ Indication will remain constant but will increase in a climb.
- ☒ Indication will rise to the top of the scale.
- ☐ Indication will drop to zero.

ID:	3405(LTS:223009, LTM:1136)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

25

If the Pitot head is blocked, what airspeed indication can he expected?

- [x] **A decrease of IAS during a climb.**
- [ ] **No change of IAS in level flight, even with large power changes.**
- [ ] **Constant IAS during a descent.**

ID: 34(LTS:222251, LTM:8)
Syllabus reference: 22. INSTRUMENTATION

Max. Points 1,00
Points 0,00

26

(Refer to Figure 85.) Of the following, which is the minimum acceptable rate of climb (feet per minute) to 9,000 feet required for the WASH2 WAGGE departure at a GS of 150 knots?

See Attachments:

1. instrument_85.jpg	figure 85	(Attachment No.1)

- ☐ 825 feet per minute.
- ☐ 1,000 feet per minute.
- ☒ 750 feet per minute.

ID:	4130(LTS:223037, LTM:1174)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

27

If the pitot line is blocked, the VSI will:

☒ **read zero ft/mm rate of climb/descent**

☐ **under-read**

☐ **read normally**

☐ **over-read**

ID:	1110(LTS:222492, LTM:248)	Max. Points	1,00
Syllabus reference:	22.2 MEASUREMENT OF AIR DATA PARAMETERS	Points	0,00

28

ASI compressibility error will increase with increase of TAS and:

☐ **Decrease with altitude.**

☒ **Increase with increase of density.**

☐ **Increase with altitude.**

ID:	813(LTS:222417, LTM:173)	Max. Points	1,00
Syllabus reference:	22.2.1 Pressure measurement	Points	0,00

29

Concerning the pitot and static system, the static pressure error:

- ☐ *is a direct effect of a blockage of the static port.*
- ☒ *is caused by disturbed airflow around the static ports.*
- ☐ *affects the alternate static port only.*
- ☐ *is a direct effect of heating of the static ports.*

ID:	1884(LTS:222719, LTM:467)	Max. Points	1,00
Syllabus reference:	22.2.1.1 Definitions	Points	1,00

30

The QFE at an aerodrome (elevation) 1790 feet is 962 hPa and the QNH 1022 hPa. If the transition level is FL040 the physical level of the transition level above the aerodrome is approximately:

☐ **2280 feet.**

☐ **2150 feet**

☒ **2464 feet**

ID: 1910(LTS:222699, LTM:476)
Syllabus reference: 22.2.4 Altimeter

Max. Points 1,00
Points 1,00

31

In an aircraft which is equipped with a CADC the combined Mach/airspeed indicator will contain ….. capsule(s).

- ☐ 1
- ☐ 3
- ☒ 2
- ☐ no

ID: 1409(LTS:222568, LTM:325)
Syllabus reference: 22.2.7 Mach meter

Max. Points 1,00
Points 0,00

32

The deviations of a direct reading magnetic compass effected only by coefficients B and C are, 3 E on 230 (C) and 4 W on 130 (C), therefore the deviation on heading 315 (C) is:

- ☐ **3,3 E**
- ☒ **3,3 W**
- ☐ **7,5 W**

ID:	1800(LTS:222682, LTM:439)	Max. Points	1,00
Syllabus reference:	22.3 MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	Points	0,00

33

**In the northern hemisphere, an aircraft takes-off on a runway with an alignment of 045°.
During the rake-off run, the compass indicates:**

☐ *a value below 045°.*

☒ *a value above 045°.*

☐ *225°.*

☐ *045°.*

ID: 1139(LTS:222497, LTM:254)
Syllabus reference: 22.3.3 Direct Reading Magnetic Compass

Max. Points 1,00
Points 0,00

34

The principle of rigidity is used for the operation of the following gyroscopic instruments:

- ☐ **Directional Gyro and Artificial Horizon.**
- ☐ **Artificial Horizon and Turn indicator.**
- ☒ **Directional Gyro and Turn indicator.**

ID:	1595(LTS:222618, LTM:375)	Max. Points	1,00
Syllabus reference:	22.4 GYROSCOPIC INSTRUMENTS	Points	0,00

35

When an aircraft is rapidly accelerated in straight and level flight, or at take-off, what inherent precession characteristic will be displayed on the attitude indicator?

- ☒ **The miniature aircraft would indicate a climb.**
- ☐ **The miniature aircraft would indicate a descent.**
- ☐ **The miniature aircraft would indicate a climb and bank.**

ID:	1459(LTS:222581, LTM:338)	Max. Points	1,00
Syllabus reference:	22.4.1 Gyroscope: basic principles	Points	0,00

36

One characteristic that a properly functioning gyro depends upon for operation is the:

- ☐ **Ability to resist precession at 90 degrees to an applied force.**
- ☒ **Position of the gyro axis relative to the Earth's axis.**
- ☐ **Resistance to deflection of the gyro rotor.**

ID:	1614(LTS:222622, LTM:379)	Max. Points	1,00
Syllabus reference:	22.4.1 Gyroscope: basic principles	Points	0,00

37

During a stabilised climbing turn at a constant rate, the instruments which indicate the correct pitch and bank are the:

- ☒ altimeter and turn-and-slip indicator;
- ☐ attitude indicator and turn-and-slip indicator.
- ☐ vertical-speed indicator and turn-and-slip indicator;

ID:	1711(LTS:222662, LTM:410)	Max. Points	1,00
Syllabus reference:	22.4.2 Rate of turn and slip indicator/Turn Co-ordinator	Points	0,00

38

A gyromagnetic compass consists of:

1 - a horizontal axis gyro
2 - a vertical axis gyro
3 - an earth's magnetic field detector
4 - an erection mechanism to maintain the gyro axis horizontal
5 - a torque motor to make the gyro precess in azimuth

The combination that regroups all of the correct statements is:

- ☐ ***1, 4.***
- ☐ ***2, 3.***
- ☒ ***2, 3, 5.***
- ☐ ***1, 3, 4, 5.***

ID:	226(LTS:222293, LTM:49)	Max. Points	1,00
Syllabus reference:	22.4.5 Remote reading compass systems	Points	0,00

39

The parameters taken into account by the flight director computer in the altitude hold mode (ALT HOLD) are:

1 - altitude deviation
2 - engine rpm
3 - ground speed
4 - pitch attitude

The combination that regroups all of the correct statements is:

- ☐ ***1, 2.***
- ☐ ***1, 3.***
- ☐ ***1.***
- ☒ ***1, 4.***

ID:	884(LTS:222433, LTM:189)	Max. Points	1,00
Syllabus reference:	22.6.3 Flight Director : design and operation.	Points	1,00

40

The purpose of a trim tab (device) is to:

- ☐ **trim the aeroplane at low airspeed.**
- ☒ **reduce or to cancel control forces.**
- ☐ **trim the aeroplane during normal flight.**
- ☐ **lower manoeuvring control forces.**

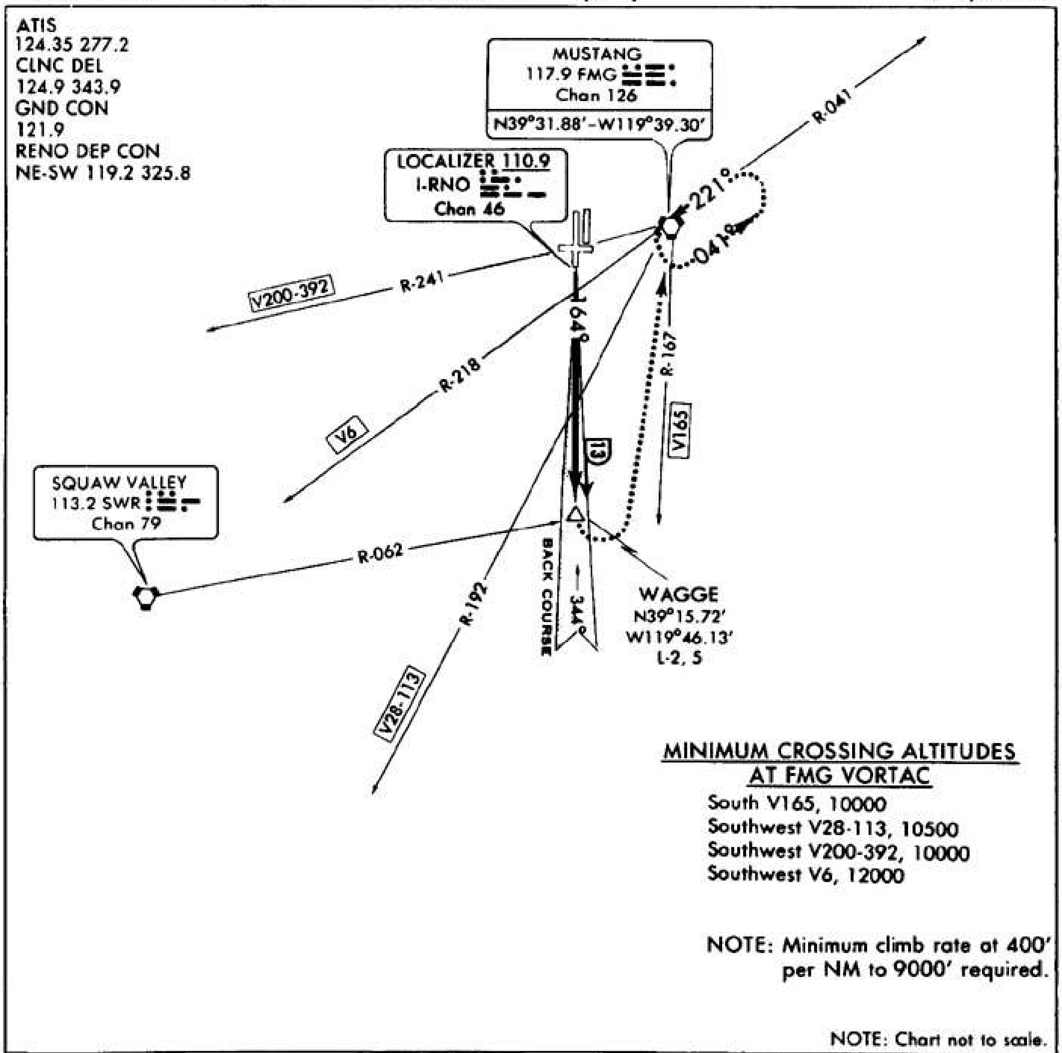
ID:	852(LTS:222427, LTM:183)	Max. Points	1,00
Syllabus reference:	22.8.1 Trim systems : design and operation.	Points	1,00

Attachment No. 1

Questions 26
figure 85
instrument_85.jpg

(VECTOR) (WASH2.WAGGE) 91094
WASHOE TWO DEPARTURE SL-346 (FAA)

RENO CANNON INTL
RENO, NEVADA



DEPARTURE ROUTE DESCRIPTION

TAKE-OFF RUNWAYS 16L/R: Climb via I-RNO Localizer south course to WAGGE INT then via radar vectors to assigned route.
LOST COMMUNICATIONS: If not in contact with departure control within one minute after takeoff, or if communications are lost before reaching 9000′, continue climb via I-RNO localizer south course to WAGGE INT, turn left, proceed direct FMG VORTAC. Cross FMG VORTAC at or above MCA, thence via assigned route or climb in holding pattern northeast on FMG R-041, left turns to cross FMG VORTAC at or above MCA for assigned route.

WASHOE TWO DEPARTURE
(VECTOR) (WASH2.WAGGE)

RENO, NEVADA
RENO CANNON INTL

Figure 85. WASHOE TWO DEPARTURE © ASA